

SIH 2026 Presentation Content: Landslide Early Warning System

[!NOTE] Below is the exact content structured for your 6-slide SIH presentation. As per the instructions in your template, the content is kept precise, uses bullet points instead of paragraphs, and fits within the strict 6-slide limit.

Tip: For Slide 3, use the provided text flow to create a visual flowchart in your actual PPT.

Slide 1: TITLE PAGE

- **Problem Statement ID:** [Insert ID from SIH Portal]
- **Problem Statement Title:** AI-powered early warning and monitoring platform for landslide-prone areas in the North Eastern Region
- **Theme:** Disaster Management / Smart Automation
- **PS Category:** Software
- **Team ID:** [Insert your Team ID]
- **Team Name (Registered on portal):** [Insert your Team Name]

Slide 2: IDEA TITLE & PROPOSED SOLUTION

Idea Title: TerraGuard NER: AI-Driven Proactive Landslide Predictor

Proposed Solution:

- **Detailed Explanation:** A cloud-integrated GIS platform analyzing multi-modal data (IMD weather, satellite, sensors) using ML to predict landslide probabilities. Features a central dashboard for authorities and an offline-capable mobile app for citizen reporting.
- **How it addresses the problem:** Shifts disaster management from reactive to proactive. Replaces manual monitoring with real-time risk heatmaps and automated, location-specific early warnings.
- **Innovation & Uniqueness:**
 - **Hybrid Data Fusion:** Combines authoritative data (IMD) with crowdsourced citizen reports (geo-tagged media).
 - **Resilient Architecture:** Low-network/offline sync capabilities tailored specifically for remote NER topographies.
 - **Hyper-Local Alerts:** Multilingual SMS & app notifications prioritized by risk severity.

Slide 3: TECHNICAL APPROACH

Technologies to be used:

- **Frontend Dashboard:** React.js (Vite), TypeScript, Leaflet (Geospatial Mapping), Recharts.
- **Mobile App:** React Native (Expo) with custom Offline Risk Engine, Turf.js, Async Storage.
- **Backend Server:** Java Spring Boot (Data JPA, WebFlux), PostgreSQL, MySQL.
- **AI/ML Service:** Python, FastAPI, XGBoost, Scikit-learn, Shap.

Methodology and process for implementation (Flow Charts):

flowchart TD

%% Define styles

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classDef server fill:#e8f5e9,stroke:#388e3c,stroke-width:2px;

classDef ml fill:#f3e5f5,stroke:#7b1fa2,stroke-width:2px;

classDef data fill:#fff3e0,stroke:#f57c00,stroke-width:2px;

%% Client Layer

subgraph ClientLayer [Client Layer]

Web[Admin Web Dashboard
React + Leaflet]:::client

Mobile[Citizen Mobile App
React Native + Offline Engine]:::client

end

%% Backend Layer

subgraph CoreServices [Core Services]

API[Java Spring Boot API Gateway]:::server

DB[(PostgreSQL / MySQL
GIS & Relational Data)]:::server

end

%% ML Layer

subgraph IntelligenceEngine [Intelligence Engine]

ML[Python FastAPI Service
XGBoost / Scikit-learn]:::ml

end

%% External Data Sources

subgraph DataSources [Data Sources]

IMD[IMD Weather APIs]:::data

ISRO[Satellite Data]:::data

end

%% Connections

Mobile -- "Syncs Geo-tagged Reports
(Offline fallback)" --> API

Web -- "Fetches Heatmaps & Alerts" --> API

API <--> DB

API -- "Requests Predictions" --> ML

IMD -. "Real-time Rainfall" .-> ML

ISRO -. "Terrain & Moisture" .-> ML

ML -- "Outputs Risk Scores" --> DB

Slide 4: FEASIBILITY AND VIABILITY

Analysis of the feasibility of the idea:

- High technical feasibility leveraging existing open government data (IMD, GSI) and mature open-source ML frameworks.
- Cloud-native architecture ensures immediate scalability across all NER states.

Potential Challenges & Risks:

- Frequent internet blackouts in remote mountainous areas.
- False positives in ML predictions causing panic.

- Spam or inaccurate citizen-reported data.

Strategies for overcoming these challenges:

- **Offline-First App:** Implemented local data caching and SMS-based fallback for critical alerts when 4G/5G fails.
 - **Human-in-the-Loop:** Dashboard highlights risks, but allows authorities to verify before broadcasting mass alerts.
 - **AI Media Verification:** Basic ML filtering to flag irrelevant images uploaded by users before they reach the dashboard.
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Slide 5: IMPACT AND BENEFITS

Potential impact on the target audience:

- **Authorities (SDMA/NDMA):** Gain a real-time, bird's-eye view for proactive resource deployment and emergency prioritization.
- **Local Communities:** Receive critical lead-time to evacuate, protecting lives and livestock.

Benefits of the solution (social, economic, environmental, etc.):

- **Social:** Drastically reduces loss of life and prevents long-term isolation of remote villages through rapid road clearance prioritization.
 - **Economic:** Minimizes infrastructure damage costs; optimizes the logistics of disaster relief funds.
 - **Environmental:** Helps in planning sustainable, climate-resilient infrastructure by identifying chronically fragile terrain and preventing over-development in high-risk zones.
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Slide 6: RESEARCH AND REFERENCES

- **Details / Links of the reference and research work:**
 - Indian Meteorological Department (IMD) API - Real-time rainfall and weather forecasts: <https://mausam.imd.gov.in/>
 - Geological Survey of India (GSI) - National Landslide Susceptibility Mapping (NLSM) data: <https://bhuvan-app1.nrsc.gov.in/gsi/>
 - NDMA Guidelines on Management of Landslides and Snow Avalanches: <https://ndma.gov.in/>
- **References:**
 - NDMA Guidelines on Management of Landslides and Snow Avalanches.
 - Research Papers on "Machine Learning applications in Landslide Susceptibility Mapping in the Himalayas."
 - Bhuvan ISRO Geo-platform for satellite imagery and terrain datasets.